

# A Representation-Degree Obstruction to Approximating Spheres by Finite-Group Orbits

Candidate partial result for Problem 2.6 of arXiv:1212.5351

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**Status.** Candidate substantial partial theorem, likely valid, pending human review. It rules out all actions with real irreducible degree below the sphere dimension; high-degree irreducible  $p$ -group actions remain open.

## 1 Source problem and scope

B. Bukh and R. Karasev, *Suborbits in Knaster's problem*, arXiv:1212.5351 [1], prove that a unit sphere of a three-dimensional subspace cannot lie in a fixed small neighborhood of an orbit of an abelian group. Their Problem 2.6 (PDF page 4) asks whether the symmetric-group construction that approximates spheres by suborbits can be modified to use a finite  $p$ -group. They point in particular to a 2-group generated by sign changes and a Sylow-2 permutation group.

*Remark 2.4.* It follows that sufficiently dense subsets of  $\mathbb{S}^2$  are not spherical suborbits of commutative group actions and cannot be approximated by spherical suborbits of commutative group actions. Indeed, if a finite set  $X \subset \mathbb{S}^2$  is such that  $\mathbb{S}^2$  is in the  $\varepsilon$ -neighborhood of  $X$  and  $X$  is in the  $\varepsilon$ -neighborhood of an orbit  $Gv$  (for an abelian  $G$  after some isometric inclusion  $\mathbb{S}^2 \subset \mathbb{R}^N$ ) then  $\mathbb{S}^2$  is in the  $2\varepsilon$ -neighborhood of this orbit, which contradicts Theorem 2.2

*Remark 2.5.* In [21] Proposition 2.3] it is proved that any  $\mathbb{S}^k$  can be  $\varepsilon$ -approximated by a suborbit of the standard coordinate-wise action of the symmetric group  $\Sigma_n$  on  $\mathbb{S}^{n-1}$  with sufficiently large  $n = n(k, \varepsilon)$ .

*Problem 2.6.* What about suborbits of  $p$ -groups in Problem 2.1? Can the proof in [21] be modified to produce a  $p$ -group instead of  $\Sigma_n$ ? Note that in [5] the following result was established during the proof of [5] Theorem 1]: There exists  $n = 2^\ell$  depending on  $k$  and  $d$ , and a  $k$ -dimensional subspace  $L \subseteq \mathbb{R}^n$  with the following property: any  $G$ -invariant polynomial  $P: \mathbb{R}^n \rightarrow \mathbb{R}$  of degree at most  $d$  is constant on the unit sphere of  $L$ . Here  $G$  is  $\mathbb{Z}_2^n \rtimes \Sigma_n^{(2)}$ , the 2-group generated by reflections and 2-Sylow permutations of coordinates. Does it follow that such  $S(L)$  is close to a  $G$ -orbit in  $\mathbb{R}^n$ ?

Figure 1: The comparison and Problem 2.6 in the source PDF, page 4.

The result below does not settle high-dimensional irreducible  $p$ -group actions. It proves that those blocks are necessary, quantitatively: a close covering orbit must place almost all of the target plane inside constituents of real degree at least the plane dimension.

## 2 Proof intuition

Decompose the representation into irreducible orthogonal blocks. Along one group orbit, the norm in each block is fixed. If projection from the target  $k$ -plane into a block has a kernel, move

a random unit vector of the plane along a kernel direction. Its projected norm acquires the nonconstant factor  $\sqrt{1-t^2}$ . No fixed orbit radius can track this factor: even its best constant approximation leaves a definite relative variance  $\delta_k$ .

Sum this unavoidable error over all noninjective blocks. When every irreducible block has dimension less than  $k$ , every projection has a kernel and the block energies sum to one. With larger constituents present, the same calculation says exactly how much of the target plane must concentrate there if the approximation error is small.

### 3 The scalar variance

For  $k \geq 2$ , let  $T_k$  be the first coordinate of a uniform point on  $S^{k-1}$ . Its density on  $[-1, 1]$  is proportional to  $(1-t^2)^{(k-3)/2}$ . Define

$$m_k = \mathbb{E}\sqrt{1-T_k^2} = \frac{\Gamma(k/2)^2}{\Gamma((k-1)/2)\Gamma((k+1)/2)}, \quad \delta_k = 1 - \frac{k}{k-1}m_k^2. \quad (1)$$

Since  $\sqrt{1-T_k^2}$  is nonconstant,  $\delta_k > 0$ .

**Lemma 1.** *For every  $R, a \geq 0$ ,*

$$\mathbb{E}(R\sqrt{1-T_k^2} - a)^2 \geq \delta_k \mathbb{E}(R^2(1-T_k^2)).$$

*Proof.* Put  $Q = \sqrt{1-T_k^2}$ . Minimizing over all real  $a$  only makes the left-hand side smaller, and the minimum is  $R^2 \text{Var}(Q)$ . Also  $\mathbb{E}Q^2 = (k-1)/k$ . Thus

$$\frac{\text{Var}(Q)}{\mathbb{E}Q^2} = 1 - \frac{(\mathbb{E}Q)^2}{\mathbb{E}Q^2} = \delta_k.$$

The beta-integral ratio for  $\mathbb{E}Q$  is exactly the gamma quotient in (1). □

### 4 The block obstruction

Let a finite group  $G$  act orthogonally on a finite-dimensional real inner product space  $V$ . Fix an orthogonal decomposition into irreducible invariant subspaces

$$V = V_1 \oplus \cdots \oplus V_m,$$

and write  $P_j$  for the projection onto  $V_j$ . Let  $L \subset V$  have dimension  $k \geq 2$ , let  $S(L)$  be its unit sphere, and set

$$J_0 = \{j : P_j|_L \text{ is not injective}\}, \quad \alpha(L) = \int_{S(L)} \sum_{j \in J_0} \|P_j v\|^2 d\sigma_L(v).$$

**Theorem 1.** *For every  $c \in V$ ,*

$$\sup_{v \in S(L)} \text{dist}(v, Gc) \geq \sqrt{\delta_k \alpha(L)}. \quad (2)$$

*In particular, if every real irreducible constituent has dimension less than  $k$ , then*

$$\sup_{v \in S(L)} \text{dist}(v, Gc) \geq \eta_k := \sqrt{\delta_k}. \quad (3)$$

*Proof.* For each  $j$ , put  $a_j = \|P_j c\|$ . Since  $V_j$  and its orthogonal complement are invariant,  $P_j$  commutes with the action and  $\|P_j(gc)\| = a_j$  for every  $g \in G$ . Therefore, for  $v \in S(L)$ ,

$$\text{dist}(v, Gc)^2 \geq \sum_{j=1}^m (\|P_j v\| - a_j)^2. \quad (4)$$

Fix  $j \in J_0$  and choose a unit  $e_j \in L \cap \ker P_j$ . A uniform vector  $v \in S(L)$  can be sampled as

$$v = T_k e_j + \sqrt{1 - T_k^2} u,$$

where  $u$  is uniform on the unit sphere of  $L \cap e_j^\perp$  and is independent of  $T_k$ . Hence

$$\|P_j v\| = \sqrt{1 - T_k^2} \|P_j u\|.$$

Condition on  $u$  and apply Lemma 1 with  $R = \|P_j u\|$  and  $a = a_j$ . Averaging in  $u$  gives

$$\int_{S(L)} (\|P_j v\| - a_j)^2 d\sigma_L(v) \geq \delta_k \int_{S(L)} \|P_j v\|^2 d\sigma_L(v).$$

Integrate (4), retain only  $j \in J_0$ , and sum. The resulting mean lower bound is  $\delta_k \alpha(L)$ , so some  $v$  satisfies (2).

If  $\dim V_j < k$ , the map  $P_j|_L$  cannot be injective. Thus if all constituents have dimension less than  $k$ , then  $J_0 = \{1, \dots, m\}$  and  $\alpha(L) = 1$  by orthogonality. This proves (3).  $\square$

## 5 Necessary concentration in large irreducibles

Let  $V_{<k}$  be the sum of all real irreducible blocks of dimension less than  $k$ , let  $Q_{<k}$  be its orthogonal projection, and let  $P_L$  denote orthogonal projection onto  $L$ .

**Corollary 1.** *If  $S(L) \subset (Gc)_\varepsilon$ , then*

$$\frac{1}{k} \text{Tr}(P_L Q_{<k}) \leq \frac{\varepsilon^2}{\delta_k}. \quad (5)$$

*Thus for  $\varepsilon \ll \eta_k$ , almost all mean-square energy of  $L$  lies in real irreducible constituents of dimension at least  $k$ .*

*Proof.* Every block in  $V_{<k}$  belongs to  $J_0$ , and spherical averaging gives

$$\int_{S(L)} \|Q_{<k} v\|^2 d\sigma_L(v) = \frac{1}{k} \text{Tr}(P_L Q_{<k}).$$

Theorem 1 therefore bounds the square of the covering radius below by  $\delta_k \text{Tr}(P_L Q_{<k})/k$ . Rearrangement proves (5).  $\square$

For  $k = 3$ ,  $m_3 = \pi/4$  and

$$\eta_3 = \sqrt{1 - \frac{3\pi^2}{32}} \approx 0.273358 > \frac{1}{4}.$$

Hence the result recovers and slightly sharpens the source's obstruction for abelian actions, while also covering every finite-group representation with real irreducible degree at most two (for example, dihedral 2-groups).

## 6 Limitations, verification, and novelty

High-dimensional irreducible  $p$ -group constituents are the exact unresolved case for this method:  $P_j|_L$  can be injective and can even have constant norm on  $S(L)$ , leaving no radial fluctuation. Complex irreducibles of finite  $p$ -groups admit monomial models, suggesting a structured permutation-phase problem, but replacing the full symmetric matching in the known construction by a Sylow-wreath action was not achieved. The packet makes no claim that the full  $p$ -group problem has a negative answer.

The gamma constants were checked numerically for  $2 \leq k \leq 10$ , including the closed form for  $k = 3$ ; code is included. The proof itself is exact and does not depend on computation.

The local run indexes and a bounded external search on 11 August 2026 checked arXiv id 1212.5351, the exact wording of Problem 2.6, weak-Knaster and  $p$ -group/suborbit variants, and the source's citation graph. No later resolution or statement of this representation-degree obstruction was found. This was not an exhaustive review, so novelty remains pending specialist checking.

## References

- [1] B. Bukh and R. Karasev, Suborbits in Knaster's problem, *Bull. Lond. Math. Soc.* **46** (2014), 269–278; arXiv:1212.5351.